Business Requirements Document (BRD)

Project 7 : AI Subscription & Auto-Debit Intelligence System

1. Executive Summary

Customers frequently subscribe to multiple services such as OTT platforms, utilities, and SaaS products. These subscriptions often operate on auto-debit systems.

However, users face issues like:

- forgetting active subscriptions
- unexpected deductions
- insufficient balance during auto-debit

The proposed solution is to build an **AI-powered Subscription Intelligence System** that:

- detects subscription transactions using NLP
- predicts upcoming debits using ML
- identifies risk of failure
- generates proactive alerts using GenAI

This system enhances **customer experience, financial planning, and payment success rate**.

2. Business Problem

Customers lack visibility into recurring payments.

Key challenges:

- subscription transactions are hidden in transaction history
- no prediction of upcoming debits
- auto-debits fail due to low balance
- users are unaware of unnecessary subscriptions

Impact:

- failed transactions
- penalty charges
- poor customer experience
- revenue loss for merchants

3. Business Objectives

1. Detect subscription transactions automatically
2. Predict upcoming auto-debit events
3. Identify potential debit failures
4. Provide proactive alerts to users
5. Improve auto-debit success rate
6. Enhance financial awareness

4. Project Scope

In Scope

- Transaction dataset analysis
- NLP-based subscription detection
- ML-based debit prediction
- Risk scoring
- GenAI alert generation
- UI dashboard

Out of Scope

- Real-time bank integration
- Payment execution
- External subscription APIs

5. Stakeholders

Stakeholder	Role
Customers	End users
Banking Team	Improve user experience
Product Team	Design features
Data Science Team	Build models

6. User Personas

Retail Customer

- Uses multiple subscriptions
- Wants alerts before deductions

Banking Analyst

- Monitors auto-debit success
- Improves customer retention

7. Functional Requirements

FR1 – Dataset Input

System must accept transaction dataset.

Fields:

- TransactionID
- Date
- Description
- Amount
- Balance
- Merchant

FR2 – Data Processing

- Clean data
- Normalize text
- Handle missing values

FR3 – NLP Subscription Detection

System must identify recurring transactions.

Examples:

- Netflix
- Spotify
- Electricity bill

Techniques:

- Keyword extraction
- Text classification

Output:

Transaction Type: Subscription
Merchant: Netflix

FR4 – Recurring Pattern Detection

System must detect:

- frequency (monthly, weekly)
- recurring dates

FR5 – Debit Prediction (ML Model)

System must predict:

Next Debit Date
Next Debit Amount

Models:

- Time series
- Regression

FR6 – Risk Prediction Model

System must predict failure risk.

Example:

Risk Score: 0.78
Reason: Low account balance

FR7 – GenAI Alert Generator

System must generate alerts.

Example:

Upcoming Debit: Netflix ₹499
Risk: Low balance
Suggestion: Add funds before 2 days

FR8 – Insights Generation

System must generate insights:

You have 5 active subscriptions.
Total monthly spend: ₹2,500

FR9 – Dashboard

Display:

- active subscriptions
- upcoming debits
- risk alerts
- monthly subscription spend

8. System Architecture

Transaction Dataset
↓
Data Cleaning
↓
NLP Subscription Detection
↓
Pattern Detection
↓
ML Prediction Models
↓
Risk Scoring
↓
GenAI Alert Generator
↓
Dashboard

9. Technology Stack

Component	Technology
Programming	Python
Data Processing	Pandas
NLP	spaCy / HuggingFace
ML	Scikit-learn
Time Series	Prophet / ARIMA
GenAI	OpenAI / Ollama
UI	Streamlit

10. Non-Functional Requirements

Requirement	Description
Performance	<10 sec processing
Local Execution	Required
Usability	Simple UI
Accuracy	High detection accuracy

11. Data Requirements

Dataset size:

50K – 200K transactions

12. Data Generation & Business Simulation

Objective

Simulate real customer transaction behavior.

Required Fields

- CustomerID
- Transaction Date
- Description
- Amount
- Balance
- Merchant
- Subscription Flag

Business Logic

- Monthly recurring payments
- Fixed subscription amounts
- Balance fluctuations
- Missed payments

Tools

- Python
- Faker

Deliverables

- Dataset
- Script
- Logic explanation

13. Success Metrics

Metric	Target
Subscription detection accuracy	>90%
Prediction accuracy	High
Risk prediction accuracy	>85%
User satisfaction	>80%

14. Risks & Mitigation

Risk	Mitigation
Incorrect detection	Improve NLP model
Irregular patterns	Advanced feature engineering
Data noise	Cleaning

15. Deliverables

- NLP subscription detection model
- ML prediction model
- Risk scoring system
- GenAI alert module
- Dashboard
- Dataset generator
- Documentation

16. Timeline

Phase	Duration
Analysis	2 days
Design	2 days
Development	1 days
Testing	1 days
Demo	1 day

Total: **7 days**